

A Surface-Content Decomposition of AI-Generated Text Detection Benchmarks

Anonymous Author(s)

Affiliation withheld for double-blind review

Abstract—A detector scoring 99% on a machine-generated-text benchmark may have learned how machine language differs from human language, or how one corpus was punctuated, and the headline number does not separate the two. We decompose the signal into three arms. A *surface-only* model reads 47 orthographic features and never sees a word, a *content-only* model reads text stripped of punctuation, casing and non-ASCII characters, and a *full* model reads text as the reference. On HC3 the two arms are indistinguishable under a paired test, 3.20% against 3.26% error, and the null holds on five of five partitions with the sign changing between them. DeBERTa reaches 0.28% on the same split, so the parity joins two weak arms rather than a ceiling. On DAIGT V2 content is stronger by a factor of 7.9 in error. Subgroup splits localise that result rather than confirm it. The surface arm reaches 0.01% error on the Reddit sub-corpus, 74.8% of HC3, and loses to content on both other domains large enough to test. Removing the cue responsible, the rate of spaces before punctuation, costs that arm 10.03 points. BERT emits identical token identifiers for strings differing only by that cue, cannot represent it, and still reaches 0.9916 weighted F1. Surface form is informative on both corpora. We claim only that it reaches parity with content on one, and there for one reason.

Index Terms—AI-generated text detection, benchmark evaluation, surface features, shortcut learning, tokenisation, SHAP, paired significance testing

I. INTRODUCTION

Detectors of machine-generated text report accuracy at or above 97% on the benchmarks the field uses most, and frequently above 99% [1], [2], [3], [4]. Read directly, those numbers say the task is close to solved on the domains these corpora represent. Read carefully, they say something weaker, because a benchmark score measures the separability of a particular corpus and not the capability a reader infers from it.

The difference matters because separability has more than one source. A detector can reach a high score by modelling how machine-generated language differs from human language, or by modelling how one corpus was assembled, punctuated and encoded. Both produce the same headline figure. Individual instances of the second have been documented, including a whitespace convention in HC3 [5], a length confound in the same corpus [6], punctuation inconsistencies requiring standardisation before comparison [7], and prompt-specific collection shortcuts [8]. Each of those is a report of one artefact. What is missing is a way to ask how much, in total, a given benchmark’s separability owes to surface form rather than content, and to place two benchmarks on the same axis so they can be compared.

This paper proposes that measurement and applies it. We compare three arms on identical splits. A *surface-only* model reads 47 orthographic features and never sees a word. A *content-only* model reads text after punctuation, casing and non-ASCII characters have been stripped away. A *full* model is a fine-tuned transformer on raw text. The gap between the first two bounds how much of a benchmark a detector could pass without reading language at all. Because the first two arms share a classifier family, they are directly comparable to each other, and the third is reported as a reference point rather than as a matched arm.

Applied to DAIGT V2 and HC3 across five model families, the answer differs sharply between the two corpora. On HC3 the surface-only and content-only arms are indistinguishable, 3.20% against 3.26% test error, and a paired test over 10,732 documents cannot separate them, while on DAIGT V2 content is stronger by a factor of 7.9. Splitting each corpus by its own subgroup labels then shows the HC3 tie is not a property of the benchmark. It belongs to the Reddit sub-corpus that makes up three quarters of it, and reverses on the other domains. The same contrast appears in the model comparison, where the best classical configuration falls 0.0007 weighted F1 short of the best transformer on DAIGT V2 and short by an error-rate factor of 16 on HC3. That tie is a statement about text budget rather than architecture, since refitting the classical grid on the exact span each transformer’s tokeniser kept removes it and the transformer then leads on both corpora.

We also report two negative results, each correcting a claim an uncontrolled version of this analysis had reached. The whitespace cue that dominates discussion of HC3 is sufficient in isolation yet unnecessary in practice, since BERT’s tokeniser provably cannot represent it and BERT still reaches 0.9916 weighted F1 there. And a character-level perturbation that drives DeBERTa from 0.9972 to 0.3644 weighted F1 looks like a targeted vulnerability but is not distinguishable from one, because the same model labels uniform random strings as human in 100% of cases at 0.98 mean confidence.

Our contributions are four. The first is the decomposition itself, stated formally in Section III-D as a reusable measurement over any human-versus-machine corpus, with the length control that stops its two arms sharing a channel. The second is its application to two widely used benchmarks, with a complete eight-configuration model comparison, cross-corpus transfer over three seeds, and a per-feature SHAP attribution that locates which orthographic property carries the asymmetry. The third is a set of three controls, on tokenisa-

tion, pipeline execution and label-free inputs, each of which overturns a conclusion the uncontrolled version of this study had reached. The fourth is a paired statistical treatment, exact-binomial McNemar with a paired bootstrap interval, applied uniformly so that no comparison rests on a difference in the fourth decimal place.

II. RELATED WORK

Detection on these benchmarks. Guo et al. [1] introduce HC3 with a RoBERTa detector at 99.82 F1, establishing both the corpus and the accuracy regime the field has worked in since, though Su et al. [2] show that regime narrows to roughly 91.7% under semantic-invariant tasks. Zhou [3] reaches competitive DAIGT V2 accuracy with character n-grams and no transformer, consistent with our findings but reported without the surface-content separation that would explain it, and Kubrusly et al. [9] likewise report strong bag-of-words results on HC3 without isolating what those features encode. Ansary [4] and Lamsiyah et al. [10] report near-ceiling DAIGT-family results in-distribution only. Two designs sit closest to ours and differ in purpose rather than ingredients. Annepaka et al. [11] mix linguistic features with a transformer for 99.45% on HC3 and Yadagiri et al. [12] pair DeBERTa with stylometric features, both combining the two channels to raise a score where we hold them apart to measure one. Socolof et al. [13] detect with style embeddings alone, a motivation we share, though a learned embedding cannot be shown to exclude content the way a hand-built orthographic vector can. Surveys [14] catalogue detector families without asking what a corpus is separable by. The field has since converged on shared evaluations that address the coverage problem rather than the separability one. RAID [15] spans eleven generators, eight domains and eleven adversarial attacks, M4 [16] adds multilingual and multi-domain coverage, and SemEval-2024 Task 8 [17] turned both into a shared task. Those benchmarks make detectors comparable across conditions. They do not ask what any one condition is separable by, which is the axis this paper adds and the reason the measurement proposed here would apply to them unchanged.

Competition solutions on DAIGT V2. The corpus was assembled for a public competition, and the winning entry [18] and strong public ones [19], [20], [21] lean on character n-grams and ensembles of classical models rather than larger language models. That is usually read as evidence classical models suffice here. Our decomposition suggests a compatible and more actionable reading, that a corpus whose surface arm alone reaches 0.9214 rewards models good at surface statistics, so a leaderboard is partly a measurement of the corpus.

Documented artefacts in these corpora. Tian et al. [5] identify the HC3 whitespace convention, release a cleaning kit, and go further than they are usually credited for. Their appendix reports that a detector consisting of the single test for one token identifier reaches 82.12 F1 on HC3 at sentence level, above the 81.89 they quote for a fine-tuned RoBERTa. Our document-level equivalent of that rule reaches 94.23, and

the difference is sentence against document rather than a disagreement. Their result is the closest existing statement of this paper’s HC3 finding. What we add is a matched content arm to compare against, so the cue can be weighed rather than only exhibited, a per-domain split that locates it, and a tokenisation control that asks whether a given detector can represent the cue at all, which turns out to change the conclusion for BERT. Baidya et al. [6] document HC3’s length confound and control it by length-matching, the same confound our two arms share and that Section IV-F closes. Ardeshtirifard [7] standardises punctuation before comparing handcrafted features to deep models without measuring what that removes, and Park et al. [8] show HC3-trained detectors rely on prompt-specific shortcuts. Where these identify individual cues, we measure the aggregate.

Shortcut learning and dataset artefacts elsewhere. The pattern is not specific to text detection. Torralba and Efros [22] showed vision datasets are identifiable from their own images, and in natural language inference a model reading only the hypothesis reaches far above chance [23], [24], while Niven and Kao [25] traced apparent argument comprehension to lexical cues that vanish under an adversarial split, with Geirhos et al. [26] giving the general account. Our surface-only arm is a hypothesis-only baseline transferred to this task, establishing how far a model gets without the capability the benchmark is taken to measure, except that it removes a property of the text rather than a span of it. Feng et al. [27] give the caveat that governs how such a baseline may be read. A partial-input model scoring highly shows a dataset is cheatable, but one scoring poorly does not show the dataset is clean, since the restricted representation may simply fail to express an artefact that is present. We therefore read a high ε_S as uninformative rather than as evidence of a clean corpus, which is why the paper reports the surface arm as an upper bound on surface-carried separability.

Robustness and its controls. Antoun et al. [28] attack HC3-trained detectors with misspellings and homoglyphs, reporting a fall from 99.88 F1 to 33.57% accuracy, and Huang et al. [29] report comparable degradation. Neither reports a label-free control, and our results suggest such collapses cannot be separated from a generic degenerate response to unreadable input without one, a methodological gap rather than a dispute about their measurements. Lai et al. [30] show ensembling preserves in-distribution accuracy while still degrading under shift, Zhang et al. [31] reframe detection as a two-sample test, and Borile and Abrate [32] ablate roughly twenty feed-forward neurons for up to 6.9 points of out-of-distribution gain, evidence that some in-distribution performance rests on localised corpus-specific features.

Fairness of the resulting decisions. Liang et al. [33] show several deployed detectors misclassify writing by non-native English speakers as machine-generated at high rates, which bears directly on ours. A detector whose separability comes substantially from orthographic regularity will concentrate its errors on writers whose orthographic habits differ from the corpus it was fitted to. We do not measure that effect here

and note the connection because it is the most consequential reason to care which arm a benchmark’s score comes from.

Cross-corpus evaluation. Alikhanov et al. [34] merge HC3 and DAIGT V2 into one 124,195-sample corpus under a topic-held-out split, reporting 82.87% to 88.86% accuracy against the near-ceiling numbers usual in-distribution. Their design asks whether one model can serve both corpora, ours asks what each corpus is separable by, so the two sit alongside rather than replacing one another. Lu et al. [35] show detectors can be evaded by prompting alone, a threat model orthogonal to the question here.

III. METHOD

A. Corpora

DAIGT V2 [36] contains 44,868 argumentative student essays, 27,371 human-written and 17,497 machine-generated by a mixture of 2023-era systems including GPT-3.5, Llama-2, Mistral-7B and Falcon-180B, assembled for a public detection competition [37]. The human side derives from the PERSUADE corpus of student argumentative writing [38]. HC3 [1] contains question-answer pairs from five English sources, contrasting human answers with a single generator, GPT-3.5-Turbo-0301, over 85,449 rows [39]. The two make a useful pair because they differ in nearly every respect that matters here. DAIGT V2 has many generators, one genre and long documents, HC3 one generator, five source domains and short ones. Both were class-balanced by downsampling the majority class, giving 34,994 and 53,806 rows, which fixes a degenerate single-class predictor at 0.333 weighted F1 rather than something respectable and is what makes the label-free control in Section IV-I readable. Training partitions balance to within 0.002 of parity.

TABLE I
CORPUS STATISTICS AFTER BALANCING. DUPLICATION IS EXACT MATCH ON LOWERCASED, WHITESPACE-NORMALISED TEXT. LEAKAGE IS THE SHARE OF TEST DOCUMENTS WHOSE TEXT ALSO APPEARS IN TRAINING OR VALIDATION.

	DAIGT V2	HC3
Rows, as published	44,868	85,449
Rows, balanced	34,994	53,806
Train / validation / test	25,196 2,800 6,998	38,785 4,289 10,732
Duplicate rows	6 (0.01%)	6,118 (7.16%)
Duplicate groups	6	5,986
Cross-label duplicate texts	0	0
Collection artefact rows	none found	483
Test leakage, group-aware split	0 (0.00%)	0 (0.00%)
Test leakage, naive stratified	0 (0.00%)	570 (5.30%)

Table I gives the duplication audit. DAIGT V2 is essentially free of internal duplication, while HC3 carries 6,118 duplicate rows in 5,986 groups plus 483 collection artefacts, typically stored API failure messages.

B. Partitioning

That duplication is why the split is group-aware. Documents are grouped by an MD5 hash of their whitespace-normalised,

lowercased text and whole groups go to one partition, so no two documents with identical normalised content land on opposite sides of a boundary. The partition is 72/8/20 rather than 80/10/10, since a fifth goes to test first and a tenth of the remainder to validation. Measured directly, the group-aware rule leaks 0 of 10,732 HC3 test documents where a plain stratified split of the same sample leaks 570, or 5.30%, and DAIGT V2 leaks nothing either way. The split is built once and reused by every model and seed, so comparing two models is never also comparing two evaluation sets.

C. Models and configurations

Five model families are evaluated. Three are classical, Naive Bayes, logistic regression and a linear support vector machine, each run under bag-of-words and TF-IDF for six classical configurations per dataset. Earlier work at full scale reported only the winning representation per model, so we report the complete grid. Two are transformers, `bert-base-uncased` [40] and `microsoft/deberta-v3-base` [41], fine-tuned with AdamW over a grid of learning rate in $\{2e-5, 3e-5\}$, batch size in $\{16, 32\}$ and weight decay in $\{0.01, 0.1\}$, which is a sixteen-run grid per dataset, eight per model, with the operating point selected on validation weighted F1 and early stopping at patience two.

DeBERTa is the strongest encoder of its size on general language-understanding benchmarks, ahead of both BERT and RoBERTa at comparable parameter counts [41], which is reason enough to include it. Two properties matter more for this study specifically. Its disentangled attention separates content from position, the relevant inductive bias when the question is how much of a corpus is separable by form, and its SentencePiece tokeniser [42] encodes leading whitespace whereas BERT’s WordPiece does not, which makes the pair a controlled contrast on exactly the cue Section IV-H examines. Benchmark standing does not by itself predict the outcome here, since the two models are level on DAIGT V2 and separate only on HC3.

Classical models receive lowercased, stopword-filtered, lemmatised text with every non-Latin character removed. Transformers receive raw text, whitespace-normalised only, truncated to 128 tokens. The asymmetry is deliberate, since stripping punctuation and casing removes signal a pretrained transformer can use, but it also confounds any classical-versus-transformer comparison, and Section IV-B measures that rather than assuming it away.

D. The surface-content decomposition

The central instrument of this paper separates how much of a benchmark’s human-versus-machine separability is available from surface form alone and how much from content alone. Write x for a document and $y \in \{0, 1\}$ for its label, with 1 denoting machine-generated. Let $\mathcal{D}_{tr}$ and $\mathcal{D}_{te}$ be the fixed training and test partitions. Two maps are defined on the raw text. The surface map ϕ_S sends a document to $\mathbb{R}^{47}$, a vector of orthographic statistics described in Section III-F, and reads

no word identity whatsoever. The content map ϕ_C first applies the normalisation

$$\nu(x) = \text{collapse}\left(\text{map}(\text{lower}(x), \sigma)\right), \quad (1)$$

where lower lowercases, map applies the character rule σ that keeps a character when it is a lowercase Latin letter or whitespace and replaces it with a space otherwise, and collapse reduces runs of whitespace to a single space. The normalised text is then sent to raw bag-of-words counts over the training vocabulary. Punctuation, casing and non-ASCII characters cannot survive ν , so they cannot enter the content arm. Word identity never enters ϕ_S .

Each arm fits one logistic regression on the same training partition,

$$h_a = \arg \min_h \sum_{(x,y) \in \mathcal{D}_t} \ell(h(\phi_a(x)), y) + \frac{1}{2C} \|h\|_2^2, \quad (2)$$

for $a \in \{S, C\}$, with ℓ the logistic loss and C fixed at the library default for both arms. Sharing the classifier family and regularisation strength is what makes the two arms comparable, since any difference is then in what the representation carries rather than how it is fitted.

The quantity we report is the error of each arm on the held-out partition,

$$\varepsilon_a = \frac{1}{|\mathcal{D}_{te}|} \sum_{(x,y) \in \mathcal{D}_{te}} \mathbf{1}[h_a(\phi_a(x)) \neq y], \quad (3)$$

and the separability gap between them,

$$\Delta = \varepsilon_S - \varepsilon_C. \quad (4)$$

A gap near zero says orthography alone reaches the same accuracy as content alone on that corpus. A large positive gap says content carries what orthography does not. The surface arm's own error answers a more direct question, how much of the benchmark a detector could pass without reading language at all.

Two properties of Δ matter. It compares two specific arms rather than partitioning the total separability, since a richer surface featurisation would lower ε_S and a stronger content model would lower ε_C , so it is a corpus-level comparison under a fixed pair of instruments, and it is not a claim that either corpus is clean. The *full* arm is the fine-tuned transformer on raw text, differing from the other two in model family, parameter count and the amount of text it reads, so we report it as a reference point and draw no conclusion from a transformer-to-classical difference without first matching the text budget.

E. The length channel and how it is closed

One channel survives into both arms, since the surface arm reads character, word and sentence counts directly and raw bag-of-words rows sum to document length, so ϕ_C recovers it too. Length is a documented confound on both corpora [6], so we close it. On the surface side we drop the five features whose magnitude scales with document size, leaving 42 rates

and ratios, and write ϕ_S^{-L} for the reduced map. Every dropped feature has a rate twin in the same vector, so the cue itself is not removed with the length. On the content side we normalise each row to unit total and rescale by the mean training document length $\bar{n}$,

$$\phi_C^{-L}(x) = \bar{n} \cdot \frac{\phi_C(x)}{\|\phi_C(x)\|_1}. \quad (5)$$

The rescaling is not cosmetic, since rows summing to one shrink every feature by roughly $\bar{n}$, so at fixed C the effective penalty becomes far heavier and the arm collapses for reasons of scale rather than of length. Section IV-F reports both variants and reads only the rescaled one. We also fit a *length-only* arm on the three pure document-size features, so the shared channel is measured directly rather than inferred from what removing it costs the other two.

F. The surface feature set

The 47 features cover punctuation-character rates, whitespace behaviour including spaces before punctuation, casing statistics, document and sentence length, non-ASCII and emoji rates, and digit rates. Every one is a count or a rate computable in one pass over the raw string, with no model, no lexicon and no language identification, which is what makes the measurement cheap enough to run on a new corpus before committing to it. Fig. 1 shows how little they overlap, which matters twice over, since the surface arm's performance is not one cue counted 47 times and the length control removes a small identifiable part of the vector rather than most of it in disguise.

G. Metrics

We report weighted F1, which is what prior work on these corpora reports, alongside error rate, since F1 compresses at this ceiling and a change from 0.9916 to 0.9972 reads as small while the corresponding error rate falls threefold. For class c with precision P_c , recall R_c , support n_c and $N = \sum_c n_c$,

$$F1_c = \frac{2P_c R_c}{P_c + R_c}, \quad F1_w = \sum_c \frac{n_c}{N} F1_c, \quad F1_m = \frac{1}{|C|} \sum_c F1_c. \quad (6)$$

Weighted and macro F1 coincide to four decimal places in every row we report, which is the check that no result here is produced by class skew. We also report the area under the ROC curve, since a threshold-free ranking measure can disagree with a thresholded one, and on DAIGT V2 it does.

H. Paired statistics

Every transformer baseline is read from the deployed checkpoint's own record rather than from the maximum over its hyperparameter grid, because an earlier version of this analysis compared a grid maximum against a fixed configuration and reported differences that did not exist. Seed spread over three seeds is reported as a range and only for the cell it was measured on, never borrowed across datasets or model families, since fine-tuning outcomes vary materially with initialisation and data order [43], [44]. A range over three runs is not a test

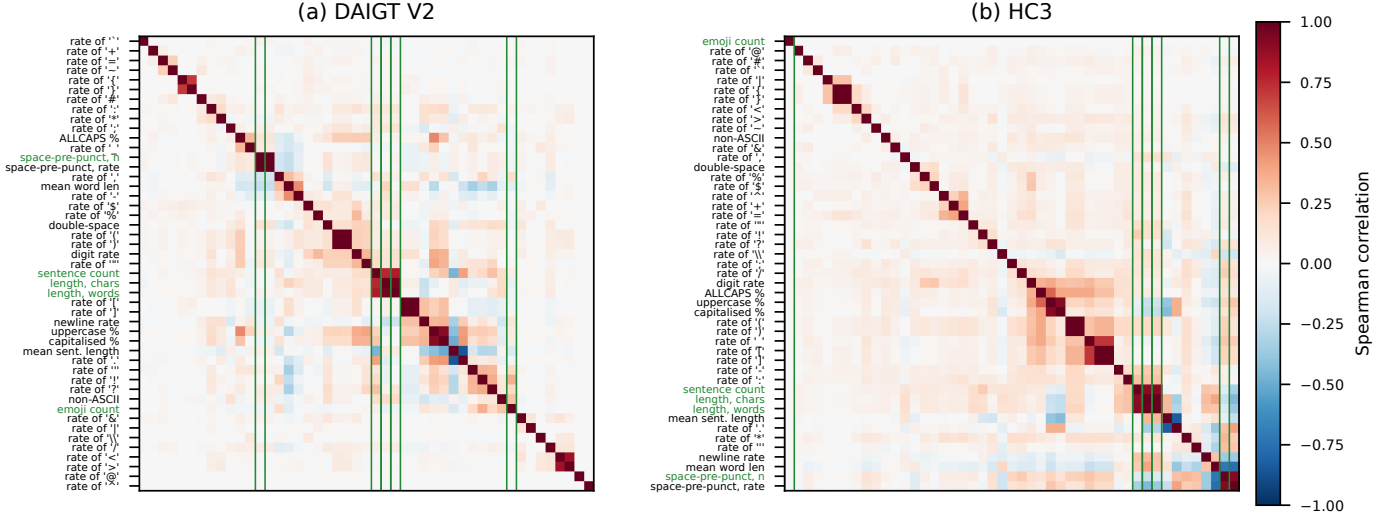


Fig. 1. Spearman correlation among the 47 surface features on each test split, ordered by average-linkage clustering on $1 - |\rho|$ so correlated blocks sit adjacent. The matrix is symmetric, so rows are labelled and columns are not. Features the length control removes are outlined in green. Features with no variance have undefined correlation and are drawn as zero.

statistic, so no comparison here rests on one, and we run no parametric tests at $n \leq 30$.

Where two models are compared on the same test set, the comparison is paired, because the two predictions are made on the same documents and treating them as independent samples discards that structure. Let b be the number of documents the first model classifies correctly and the second does not, and c the reverse. Under the null that the two have equal error, the discordant pairs split evenly, so

$$b \mid (b + c) \sim \text{Binomial}(b + c, \frac{1}{2}), \quad (7)$$

and we report the exact two-sided binomial p -value [45] rather than the chi-squared approximation, which is unreliable when $b + c$ is small, as it is for several comparisons here.

An exact test reports whether a difference is distinguishable from zero, not how large it is, so each comparison also carries a paired bootstrap interval on the error difference [46]. Writing $e_i^{(1)}$ and $e_i^{(2)}$ for the two models' per-document error indicators, we resample document indices with replacement $B = 10,000$ times and take the empirical 2.5th and 97.5th percentiles of

$$\hat{\Delta}^{(b)} = \frac{1}{n} \sum_{i \in I_b} (e_i^{(1)} - e_i^{(2)}), \quad (8)$$

where I_b is the b th resample. Resampling index sets rather than each model's errors independently is what preserves the pairing.

A bootstrap resamples one test set and therefore says nothing about how the corpus was partitioned in the first place. Because the central claim of this paper is a null on HC3, and a null is exactly the kind of claim a single lucky partition can manufacture, the whole decomposition is repeated over five group-aware partitions of the same balanced sample, varying only the split seed. Section IV-G reports what that shows.

I. Controls

Three controls guard the claims below, each added after an uncontrolled version of the corresponding experiment produced a conclusion that did not survive. The *pipeline-fires* control exists because our whitespace cleaning produces bit-identical BERT predictions on HC3, which alone is indistinguishable from a cleaning step that never ran, so we apply the same pipeline to DAIGT V2, where it also removes emoji that WordPiece represents, and confirm predictions there do change. The *tokenisation* control compares token identifier sequences for minimally different strings, since whether a detector can exploit a character-level cue is a property of its tokeniser and directly testable. The *label-free* control scores 400 inputs per condition carrying no human-versus-machine information at all, including uniform random character strings, token-shuffled text, punctuation-only strings, a repeated token and the empty string, without which a confident prediction on corrupted input is indistinguishable from a learned response to it.

IV. RESULTS

A. A classical model matches a transformer on DAIGT V2 but not on HC3

Table II reports every configuration on both datasets, and the two benchmarks part company. On DAIGT V2 the best classical configuration, an SVM over TF-IDF at 0.9910 weighted F1, sits 0.0007 below the best transformer, and on the same documents the two disagree on 99 cases split 47 to 52, so McNemar returns $p = 0.69$ and the paired interval on the error difference, $[-0.21, +0.34]$ points, contains zero. On HC3 the same comparison is 0.9551 against 0.9972, 484 discordant cases split 16 to 468, $p < 10^{-6}$, and an error difference of +4.21 points with interval $[+3.82, +4.61]$, an error-rate ratio of sixteen to one.

Fig. 2 gives the corresponding ROC curves in the region that matters for a deployed detector, where a false positive

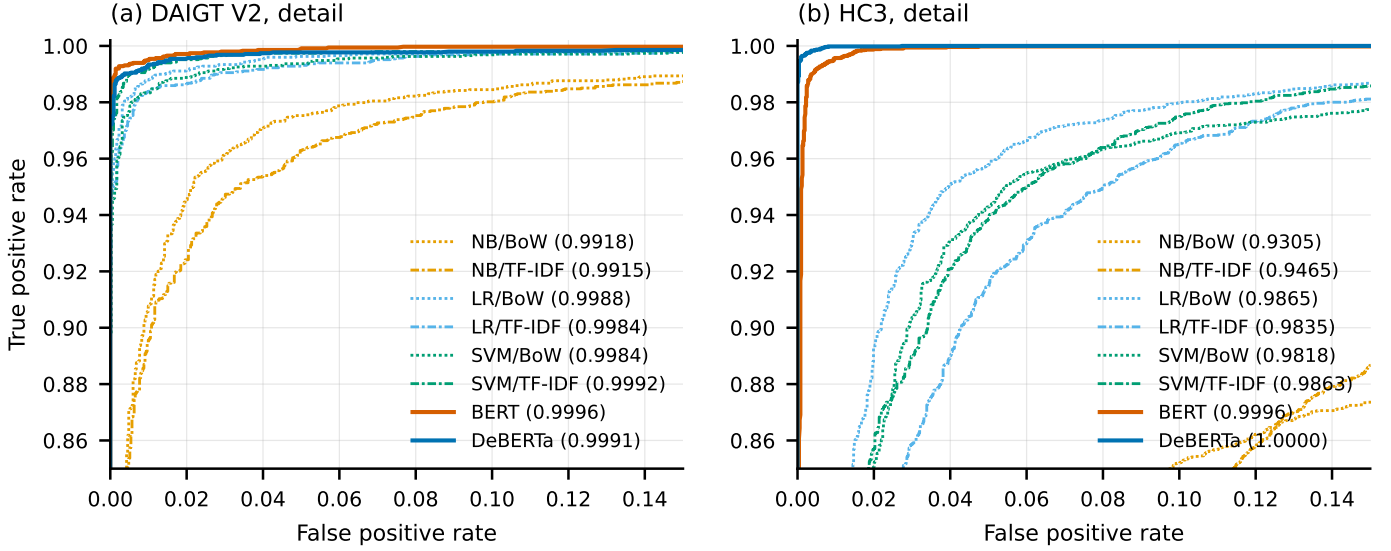


Fig. 2. ROC curves for all eight configurations, restricted to the low-false-positive region where the models differ. Areas under the full curve are given in the legend. Each series carries a distinct dash pattern, so the ordering survives greyscale.

TABLE II

ALL EIGHT CONFIGURATIONS PER CORPUS. TRANSFORMER ROWS ARE DEPLOYED CHECKPOINTS, NEVER GRID MAXIMA. ERROR RATE ACCOMPANIES F1, WHICH SATURATES HERE. BOLD MARKS THE BEST CLASSICAL AND BEST OVERALL F1 PER CORPUS.

Model	Rep.	DAIGT V2		HC3	
		F1	err %	F1	err %
Naive Bayes	BoW	0.9591	4.09	0.8713	12.87
Naive Bayes	TF-IDF	0.9577	4.23	0.8670	13.28
Logistic Reg.	BoW	0.9893	1.07	0.9551	4.49
Logistic Reg.	TF-IDF	0.9857	1.43	0.9365	6.35
SVM	BoW	0.9870	1.30	0.9475	5.25
SVM	TF-IDF	0.9910	0.90	0.9449	5.51
BERT	raw	0.9916	0.84	0.9916	0.84
DeBERTa	raw	0.9917	0.83	0.9972	0.28

is an accusation. Over the full range both corpora look alike, every curve close to the corner, and only this view separates them.

Every transformer figure above is the seed-42 checkpoint. Repeating each of the four at seeds 123 and 456 moves them very little. DAIGT V2 BERT averages 0.9921 over a range of 0.0011 and DeBERTa 0.9930 over 0.0024, so the two overlap and the tie is not a seed accident. HC3 BERT averages 0.9930 over 0.0036 and DeBERTa 0.9970 over 0.0005, and those two ranges are disjoint, so the separation that matters to this paper survives reseeding while the tie that also matters survives it in the other direction.

Ranking by area does not reproduce ranking by F1. BERT takes the higher area on DAIGT V2, 0.999641 against 0.999144, while DeBERTa takes the higher F1, and neither ordering survives contact with the predictions, which disagree on 75 documents split 37 to 38 at $p = 1.00$. On HC3 the pair separates cleanly. We report both measures because of this, and read neither ordering on DAIGT V2 as a result.

TABLE III

CLASSICAL MODELS REFITTED ON THE EXACT CHARACTER SPAN EACH TRANSFORMER’S TOKENISER KEPT, SO BOTH SIDES READ THE SAME TEXT. THE WINDOW COMES FROM OFFSET MAPPING, WITH NO DECODE ROUND-TRIP.

Data	Window	chars	best classical err %		transf.
		kept	full text	128-tok	err %
DAIGT V2	BERT	0.335	0.90	2.10	0.84
DAIGT V2	DeBERTa	0.341	0.90	2.09	0.83
HC3	BERT	0.746	4.49	5.66	0.84
HC3	DeBERTa	0.756	4.49	5.72	0.28

B. The DAIGT V2 tie is a text budget, not an architecture result

The DAIGT V2 tie is not a matched comparison. The classical models read the whole document while the transformers read 128 tokens, which is 33.5% of a mean DAIGT V2 essay against 74.6% of a mean HC3 answer, so Table III refits the classical grid on the exact span each checkpoint’s tokeniser kept. The best DAIGT V2 configuration rises from 0.90% to 2.10% error against BERT’s 0.84%, a difference of 1.257 points with interval $[+0.900, +1.629]$, and the DeBERTa window returns the same figure. On HC3 the gap widens from 4.49% to 5.66%. Held to one text budget the transformers lead on both corpora, and what the unmatched comparison showed was never about architecture but about information access, a bag-of-words model reading a whole essay against a transformer reading its opening.

C. On HC3, orthography alone matches content alone

Table IV gives the decomposition. On HC3 the two arms are statistically indistinguishable, 3.20% error from orthography against 3.26% from content, disagreeing on 625 documents split 316 to 309, so McNemar returns $p = 0.81$ and the paired bootstrap puts the difference at -0.07 points with interval $[-0.52, +0.40]$. Forty-seven features that never read a word do as well as a bag-of-words model over the whole vocabulary.

TABLE IV
THE SURFACE ARM READS 47 ORTHOGRAPHIC FEATURES AND NEVER A WORD. THE CONTENT ARM IS BAG-OF-WORDS AFTER PUNCTUATION, CASING AND NON-ASCII ARE STRIPPED. THE LOWER BLOCK CLOSES THE LENGTH CHANNEL BOTH ARMS SHARE.

Arm	DAIGT V2		HC3	
	F1	err %	F1	err %
surface-only	0.9214	7.86	0.9680	3.20
content-only	0.9901	0.99	0.9674	3.26
full transformer	0.9917	0.83	0.9972	0.28
<i>length channel closed</i>				
length-only (3 feats)	0.7540	24.59	0.8475	15.23
surface-only, no length	0.9007	9.93	0.9662	3.37
content-only, length-norm.	0.9906	0.94	0.9371	6.28

On DAIGT V2 the same two arms separate by a factor of 7.9 in error, 7.86% against 0.99%, unambiguously so, with 569 discordant documents split 44 to 525 at $p < 10^{-6}$ and a difference of +6.87 points, interval [+6.23, +7.52].

The content arm is deliberately unfiltered, and that choice is load-bearing rather than incidental. Refitting it with the stop-word removal and lemmatisation Table II’s classical pipeline uses reproduces that pipeline exactly, 1.07% error on DAIGT V2 and 4.49% on HC3, and it is the weaker content model on both. On HC3 the filtered arm loses to orthography by 1.30 points, interval [−1.79, −0.78] at $p < 10^{-6}$, where the unfiltered arm ties it. We report the unfiltered arm because it is the stronger opponent, which makes the parity claim harder to make rather than easier. It also means the sixteen-to-one error ratio quoted above is computed against the filtered pipeline, and against the stronger content model the ratio is 11.6 to one.

Surface form is informative on both benchmarks, since DAIGT V2’s surface-only arm reaches 0.9214 weighted F1, far above the 0.50 a coin gives and the 0.333 a degenerate single-class predictor gives. The finding is not that one corpus is clean and the other is not, but that only on HC3 does surface rise to parity with content. The content-only arm reaches 0.9901 on DAIGT V2 against the transformer’s 0.9917, within 0.16 points of error, under the same unmatched text budget as Section IV-B, while on HC3 DeBERTa improves on either arm by roughly 0.03 weighted F1, consistent with combining information neither arm holds alone.

D. One orthographic feature carries the HC3 result

The decomposition says how much orthography carries, not which orthographic property carries it, and the two corpora differ there as sharply as in the aggregate. Fig. 3 gives per-feature Shapley attributions [47] over the surface arm, computed exactly rather than sampled because the arm is linear.

On HC3 the single largest contributor is the rate of spaces preceding punctuation, at mean absolute SHAP 7.81 against 3.35 for the next feature, and its fitted coefficient is negative, so its presence drives the human label. That is the collection convention Tian et al. [5] identified, recovered here without being looked for and quantified against the rest of the feature set. Section IV-H then shows why its presence does not license the conclusion a reader would reach next.

On DAIGT V2 the leading contributors are document length at 2.77, mean word length at 1.88 and casing statistics, none of which exceeds 2.8 and none of which is a collection convention in the same sense. The surface arm on that corpus is reading a diffuse stylistic signal rather than a single artefact, which is consistent with its much weaker showing against the content arm.

E. The HC3 parity belongs to one sub-domain

TABLE V
HC3 BY SOURCE DOMAIN, EACH REBALANCED TO PARITY ON A GROUP-AWARE SPLIT RESTRICTED TO THAT DOMAIN. THE LAST TWO COLUMNS GIVE THE SHARE OF DOCUMENTS CARRYING A SPACE BEFORE PUNCTUATION.

Domain	<i>n</i>	surface err %	content err %	cue present in	
				human	machine
reddit_eli5	6,690	0.01	2.66	98.9%	0.3%
finance	684	7.60	3.07	5.6%	0.1%
medicine	250	4.00	0.40	17.0%	0.1%
open_qa	198	4.04	5.05	60.7%	0.4%
wiki_csai	168	5.36	11.90	1.6%	0.2%

open_qa and wiki_csai fall below 200 test rows.

Both corpora carry subgroup labels, and rejoining them to the existing splits recovers a source for every row. Running the same decomposition per sub-corpus rather than per corpus moves the HC3 result somewhere more specific.

Table V gives it. On reddit_eli5 the surface arm reaches 0.01% error, roughly one mistake in 6,690 documents, and beats content by 2.65 points. On the two other domains with enough test rows to support a claim the ordering reverses and content wins clearly, by 4.53 points on finance and 3.60 on medicine. Since reddit_eli5 is 74.8% of the balanced corpus, the corpus-level parity reported in Section IV-C is substantially that one domain’s result.

The mechanism is the cue Section IV-D identified, and it is concentrated in the same place. It appears in 98.9% of reddit_eli5 human documents against 0.3% of machine ones, and decays across the rest, 60.7% on open_qa, 17.0% on medicine, 5.6% on finance and 1.6% on wiki_csai. A single boolean rule, treating a document as human when the cue is present, scores 99.22% accuracy on reddit_eli5 and 94.23% on the whole corpus at document level. Tian et al. [5] report 82.12% for the equivalent single-token rule at sentence level, and the gap between their figure and ours is the difference between a sentence and a document rather than a disagreement.

The same treatment on DAIGT V2 pairs each machine generator against the shared human essay pool. Content wins on every generator, consistent with the corpus-level result, but the surface arm’s error spans roughly fifteenfold across them, from 0.50% to 15.54%. Two pairs are the informative ones, because they are the same underlying model contributed by different people. Mistral-7B-Instruct appears at 1.09% under one contributor and 4.44% under another, and PaLM at 2.32% and 5.38%. Surface separability on that corpus therefore tracks collection provenance more closely than it tracks which model

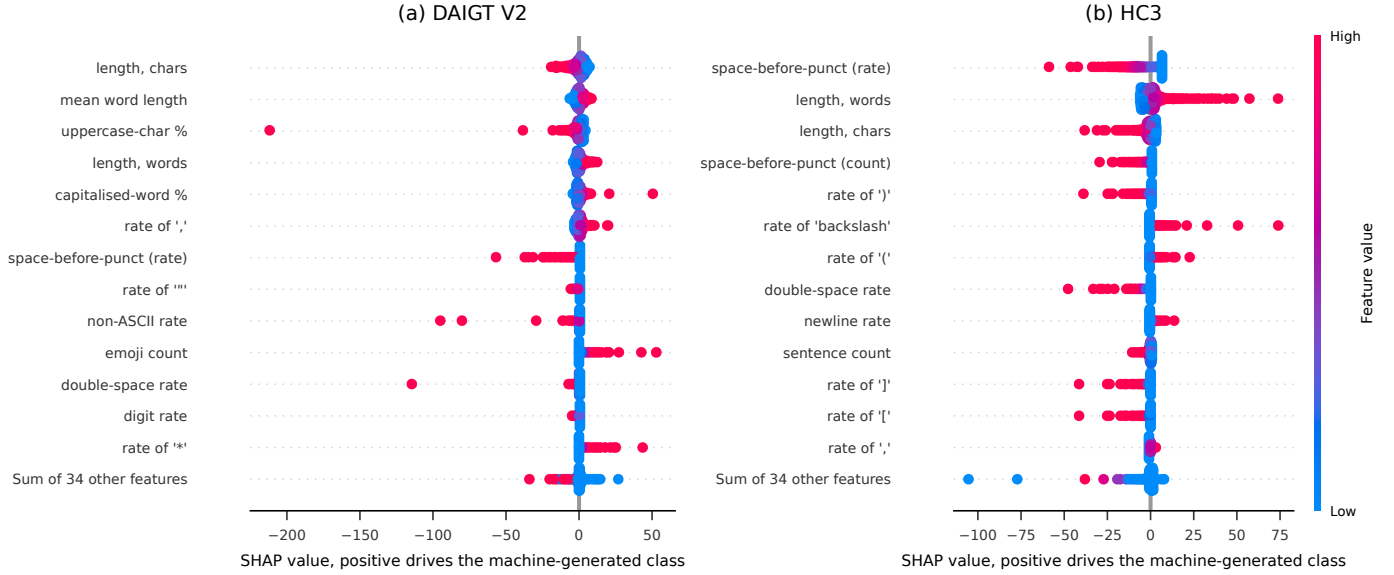


Fig. 3. SHAP attributions over the surface-only arm, computed exactly with a linear explainer on 2,000 held-out documents per corpus. Each point is one document, positioned by its contribution to the machine-generated class and coloured by feature value.

produced the text, which is the same conclusion the HC3 domain split reaches by a different route.

F. Neither result is an artefact of document length

Both arms can read document length, so the comparison above could be a length comparison in disguise. Length alone reaches 24.59% error on DAIGT V2 and 15.23% on HC3, well short of either arm but far from nothing, and closing the channel on both sides sharpens rather than weakens the two conclusions. On DAIGT V2 the length-free content arm loses nothing, 0.94% against 0.99% at $p = 0.71$, while the length-free surface arm rises to 9.93%, widening content's advantage from 7.9 to 10.6 times. On HC3 the surface arm barely moves, 3.20% to 3.37%, while the content arm rises to 6.28%, so orthography no longer merely matches content but beats it by 2.91 points, interval $[+2.37, +3.44]$. Removing length costs the content arm eighteen times what it costs the surface arm, the opposite of what a length-driven surface result would produce.

The instrument nearly produced a third false conclusion here. Normalising bag-of-words rows to unit total without restoring their scale shrinks every feature by roughly the mean document length, so at fixed regularisation the penalty becomes far heavier and the arm collapses to 9.69% on DAIGT V2 and 14.25% on HC3. That reads as a dramatic length effect and is an artefact of scale, visible in the optimiser trace, where the unrescaled arm converges in 16 to 24 iterations against 121 to 192 for the raw-count arm. Every figure above uses rows rescaled to the raw-count magnitude.

G. The HC3 null survives repartitioning

A null on one partition is exactly what a lucky split manufactures, so the decomposition runs again over five group-aware partitions of the same balanced sample. Seed 42 reproduces the single-split numbers exactly, which is the

TABLE VI
THE DECOMPOSITION OVER FIVE GROUP-AWARE PARTITIONS OF THE SAME BALANCED SAMPLE, VARYING ONLY THE SPLIT SEED. SPLIT VARIANCE RUNS 0.17 TO 1.29 PERCENTAGE POINTS.

Arm	DAIGT V2 mean err % (range)	HC3 mean err % (range)
surface-only	7.15 (6.74–7.86)	3.16 (3.07–3.33)
content-only	0.82 (0.66–0.99)	3.24 (3.06–3.41)
surface-only, no length	9.76 (9.47–9.93)	3.26 (3.14–3.37)
content-only, length-norm.	0.84 (0.77–0.94)	6.02 (5.82–6.28)
length-only	24.31 (23.56–24.85)	15.32 (15.23–15.49)

harness self-check that makes the other four worth reading. Content leads on DAIGT V2 on all five, by 6.00 to 6.87 points at $p < 10^{-6}$ throughout. On HC3 the difference reaches significance on none of them, with p of 0.81, 0.27, 0.34, 0.23 and 0.78, and it changes sign between partitions, from -0.30 to $+0.27$ points. Sign instability is the strong form of the null, since an underpowered real difference keeps its sign. The length-free comparison behaves the opposite way, significant on all five and stable in sign from -2.67 to -2.91 points, so the advantage orthography gains once length is removed is not a partition effect.

H. A model that cannot represent the cue reaches 0.9916 anyway

The most-discussed HC3 artefact is a space preceding punctuation in human answers [5], which Section IV-D confirms is the strongest orthographic feature there. It appears 10.745 times per human document against 0.013 per machine one, in 88.7% of human documents against 0.3%, and DAIGT V2 carries it far more weakly at 0.548 against 0.004 per document.

The natural next inference is that HC3-trained detectors exploit it. Tokenisation makes that testable rather than rhetorical, and the test does not support it. BERT's WordPiece tokeniser splits on punctuation irrespective of adjacent whitespace. For the strings "the answer is simple ." and

"the answer is simple." it emits identical identifier sequences, in three of three pairs tested, while DeBERTa's SentencePiece encodes the leading whitespace and distinguishes all three. BERT cannot represent the cue at all and still reaches 0.9916 weighted F1 on HC3. The cue is therefore sufficient in isolation, since the surface arm that reads it reaches 0.9680, and unnecessary in practice, since a model blind to it reaches 0.9916.

This is also why removing the cue changes nothing for BERT. Whitespace cleaning on HC3 yields predictions bit-identical to the uncleaned run, verified by SHA-256 over the saved probability arrays, which alone would be indistinguishable from a cleaning step that never executed. The same pipeline on DAIGT V2, where it also removes emoji that WordPiece represents, does change BERT's predictions, so the pipeline fires and the null is specific to whitespace.

The cue is also close to load-bearing for the surface arm, which the attribution alone does not establish. Applying the cleaning kit Tian et al. released, and nothing else, changes 44.5% of HC3 documents and moves the surface arm from 3.20% to 13.22% error, a loss of 10.03 points with interval $[-10.69, -9.38]$. The content arm is untouched at 3.26%, since the cleaning removes characters its normalisation already discards, so after cleaning orthography sits 9.96 points behind content on the corpus where it had tied. HC3's surface parity is therefore not a diffuse property of the corpus but a single collection convention, and a reader who cleans the corpus before using it should expect the surface arm to behave as it does on DAIGT V2. We note that our own cleaned artefact applies length matching alongside the whitespace fix, so this experiment rebuilds the cleaned text without it to isolate what the kit removes.

We do not attribute the BERT-to-DeBERTa difference on HC3 to this cue, since the two models differ in tokeniser, architecture, pretraining corpus and parameter count. What the experiment establishes is narrower and, for anyone planning a cleaning ablation, more useful, that a null from removing a cue is uninterpretable without first checking the model could read it.

I. The adversarial collapse is not distinguishable from a degenerate response

TABLE VII

LABEL-FREE CONTROL. SHARE OF INPUTS ASSIGNED THE *human* LABEL, WITH MEAN MAXIMUM SOFTMAX IN PARENTHESES, OVER 400 INPUTS PER CONDITION THAT CARRY NO HUMAN-VERSUS-MACHINE INFORMATION.

Checkpoint	rand. chars	shuffled	punct.
DAIGT V2 BERT	0% (0.91)	94.8% (0.99)	97.8% (0.71)
DAIGT V2 DeBERTa	100% (0.99)	100% (1.00)	100% (1.00)
HC3 BERT	51.0% (0.71)	98.5% (1.00)	100% (1.00)
HC3 DeBERTa	100% (0.98)	100% (1.00)	100% (0.99)

Perturbing HC3 test text with character-level typos at 10% of alphabetic characters drives DeBERTa from 0.9972 to 0.3644 weighted F1. The confusion matrix is one-directional, $[[5376, 1], [5207, 148]]$, with human documents still classified

almost perfectly and machine documents reassigned to human. On unperturbed data the same checkpoints split their errors roughly evenly between the two directions, so against that baseline this looks like a targeted vulnerability. It is not what the control shows.

Table VII reports the label-free control. On inputs carrying no human-versus-machine information the same model assigns the human label to uniform random character strings in 100% of cases at mean confidence 0.98, and all four checkpoints label token-shuffled and punctuation-only text human between 94.8% and 100% of the time. The behaviour is specific to neither HC3 nor the attack.

Three candidate mechanisms were tested and none survives. Majority-prior collapse is excluded by the training balance, 0.4981 against 0.5019 on DAIGT V2 and 0.4999 against 0.5001 on HC3. Cue injection is excluded because the perturbation does not create the whitespace artefact, moving the machine-side count only from 0.000 to 0.005. And a learned association from disfluency to human authorship is unsupported, since subword fertility runs the wrong way, 1.1192 for HC3 human text against 1.1980 for machine.

We therefore decline to read our own perturbation numbers as robustness measurements, since the experiment establishes a requirement on future work rather than a property of these benchmarks. One contrast within them is worth stating even so. Back-translation rewrites the text while leaving it readable and costs all four models between 0.004 and 0.020 weighted F1, whereas character-level corruption, which makes the text unreadable, produces the collapses, with HC3 BERT falling to 0.3746 and HC3 DeBERTa to 0.3644 under 10% typos. That pattern fits the label-free reading and not a targeted cue-removal one.

J. Transfer and ensembling

TABLE VIII

CROSS-CORPUS TRANSFER OVER THREE SEEDS PER CELL, EACH MODEL TRAINED ON ONE CORPUS AND EVALUATED ON THE OTHER WITHOUT ADAPTATION. RANGES ARE OBSERVED MINIMA AND MAXIMA, NOT CONFIDENCE INTERVALS.

Trained	Model	in-dom.	cross-dom. (range)	gap
DAIGT	BERT	0.9921	0.7902 (0.770–0.810)	0.2019
DAIGT	DeBERTa	0.9930	0.9096 (0.891–0.926)	0.0833
HC3	BERT	0.9930	0.8311 (0.813–0.852)	0.1619
HC3	DeBERTa	0.9970	0.8512 (0.830–0.887)	0.1458

Table VIII reports transfer in both directions over three seeds, with each model trained on one corpus and evaluated on the other without adaptation. Every cell loses between 0.0833 and 0.2019 mean weighted F1, which is what should happen if a substantial part of each corpus's separability is corpus-specific. DeBERTa's gap is roughly half BERT's in both directions at the mean level. We report that as a mean-level pattern that survives three-seed averaging rather than as a confirmed effect, because the observed ranges for BERT's two directions nearly touch and this project runs no parametric tests at three seeds.

A soft-vote ensemble over the two deployed checkpoints, its mixing weight chosen on validation and applied once to test, does not reliably beat its stronger member. On DAIGT V2 an equal mix reaches 0.9936 against DeBERTa’s 0.9917, but McNemar returns $p = 0.085$ and the interval on the error difference, $[-0.39, +0.01]$ points, contains zero. On HC3 the sweep places zero weight on BERT, so the ensemble is DeBERTa and reports DeBERTa’s score. We record that degenerate weight rather than presenting it as an ensemble result. Reading either row as an improvement would be treating a fourth-decimal difference as a finding, which is consistent with Lai et al. [30], who find ensembling preserves in-distribution accuracy without touching the shift that actually degrades these models.

K. A consolidated view

Fig. 4 collects the results above. Panel (c) is where the central claim stands or falls, with the DAIGT V2 series six to seven points above zero on every partition and the HC3 series straddling it and changing sign. Panel (f) shows the same distinction as intervals, where grey marks those containing zero.

V. DISCUSSION

The decomposition separates two benchmarks that a headline accuracy figure does not, and applied per sub-corpus it separates one benchmark from itself. On HC3 orthography alone matches content alone and a paired test cannot distinguish them on any of five partitions, while on DAIGT V2 content is stronger by a factor of 7.9 in error. Section IV-E then narrows the first of those. The tie is the Reddit sub-corpus, which carries the cue in 98.9% of its human documents and is 74.8% of the corpus, and content wins on the other domains that can support a claim. Neither a practitioner choosing an evaluation corpus nor a reviewer reading a reported score can recover that distinction from the score itself. The attribution in Section IV-D says where it comes from, since HC3’s surface arm is close to a single-feature model while DAIGT V2’s reads a diffuse mixture of length, word shape and casing. Those are different kinds of corpus even though both are separable by form, and the practical consequence differs, because a single collection convention can be cleaned and a diffuse stylistic signal cannot, and may not even be an artefact.

The measurement does not license a claim that either corpus is clean. DAIGT V2’s surface-only arm reaches 0.9214 weighted F1, so surface form is substantially informative on both, and the finding is comparative, that surface reaches parity with content on one corpus and not the other, bounded by the arms we built.

Two results began as the opposite conclusion and say more about method than about these corpora. A cleaning experiment reporting no change is uninterpretable without evidence the pipeline ran, and a tokeniser that cannot represent the cue being cleaned makes the null a fact about the model rather than the corpus. A collapse under perturbation is uninterpretable without a label-free control, since a model that calls random

strings human at 0.98 confidence produces the same curve with no vulnerability behind it. We suggest comparable published collapses be read with that control in mind.

Three reporting practices here were adopted after an error reached a draft. Transformer baselines are read from the deployed checkpoint’s own record rather than a grid maximum, after an ablation comparing the two reported differences that did not exist. Seed spread is quoted only for the cell it was measured on, after one was borrowed across corpora as a noise floor. And a leakage rate quoted from an earlier draft had no source file, so it was removed and then measured.

There is a connection to fairness we did not measure and that is the most consequential reason to care which arm a score comes from. Liang et al. [33] show deployed detectors misclassify non-native English writing as machine-generated at high rates. A detector whose separability rests on orthographic regularity should be expected to concentrate its false positives on writers whose orthographic habits differ from the corpus it was fitted to. Our measurement predicts which corpora carry that risk most sharply. Establishing the effect would need author metadata neither corpus provides.

VI. LIMITATIONS

Several limits bound every number above. The transformers see at most 128 tokens, so the DAIGT V2 transformer results describe classification from an essay’s opening, and the cross-corpus comparison in Table VIII is not like-for-like in the text each model receives. The four deployed checkpoints were run at three seeds and move by 0.0005 to 0.0036 weighted F1, but the hyperparameter grid behind them is single-seed, so a different seed could in principle have selected a different winner. The content arm is bag-of-words, so word order and syntax lie outside it and a syntactic arm could narrow the DAIGT V2 gap, while the surface set is 47 hand-built features, two with no variance on the DAIGT V2 test split and one on HC3, so ε_S is an upper bound. HC3 has one generator collected at one time and both corpora are English, so the orthographic conventions the surface arm reads are not language-independent. We evaluate two corpora and their eighteen sub-corpora across five model families and claim nothing beyond them. Five of those sub-corpora fall below 200 test rows after rebalancing and are reported as underpowered rather than as null results.

VII. CONCLUSION

We proposed a decomposition that measures how much of a machine-generated-text benchmark’s separability is available from surface form alone rather than from content, and applied it to DAIGT V2 and HC3 across five model families and eight evaluated model configurations per corpus. On HC3 the two arms are indistinguishable under paired testing on five partitions, and attribution, ablation and a per-domain split agree that a single collection convention in a single dominant sub-corpus accounts for most of the surface arm. On DAIGT V2 content leads by a factor of 7.9 in error and the surface signal is diffuse. Closing the length channel both arms share

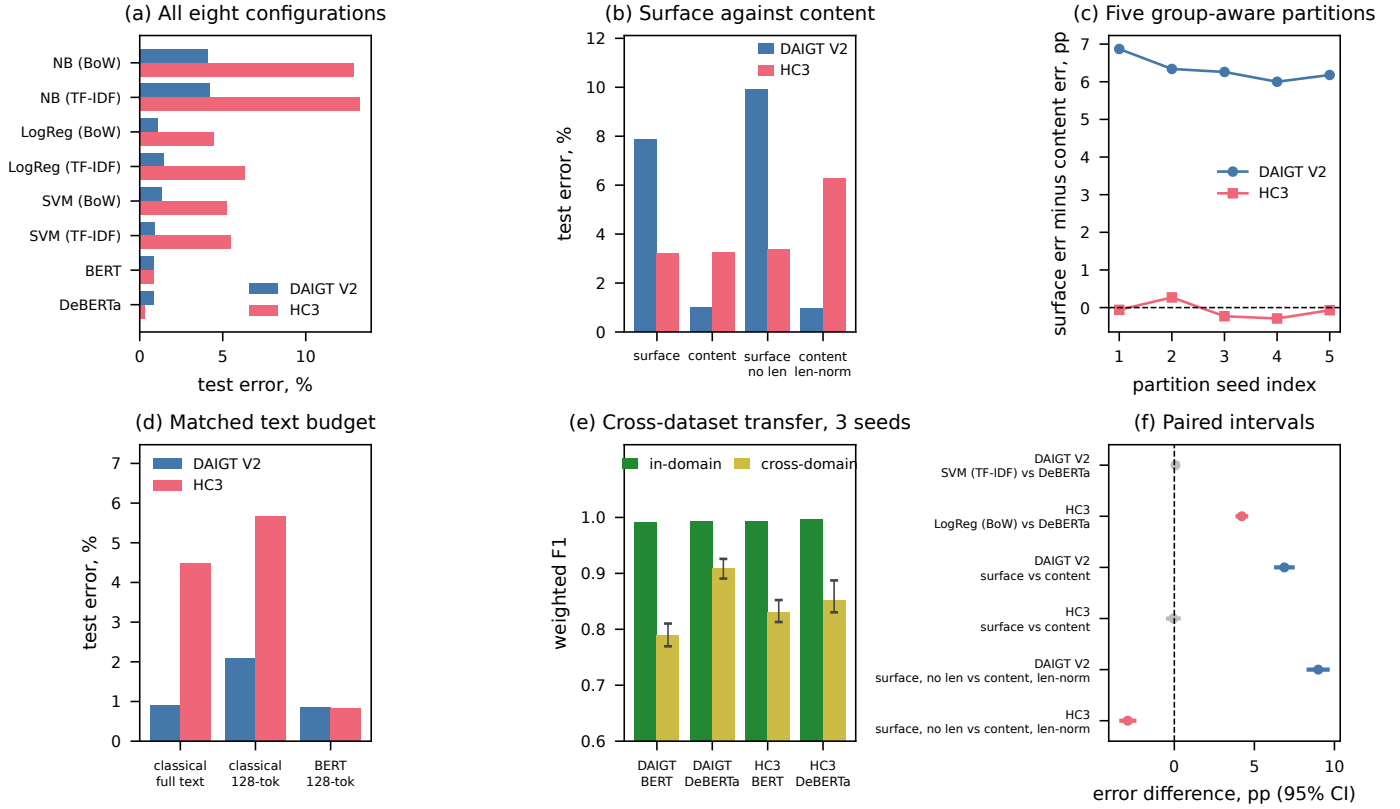


Fig. 4. Every headline result in one view. Panels give the model ranking (a), the decomposition with and without the length channel (b), the surface-minus-content difference across five partitions (c), the matched text budget (d), cross-corpus transfer (e), and the paired intervals behind each significance claim (f), where grey marks an interval containing zero.

strengthens both conclusions. Three controls, on tokenisation, pipeline execution and label-free inputs, each overturned a conclusion this study had reached without them.

The measurement itself is the portable part. It needs no new annotation, it fits two logistic regressions, and it runs in minutes on a laptop. It yields one number per corpus that a headline accuracy cannot express, namely how much of that corpus a model could pass without reading the language. We would encourage reporting it alongside any new detection benchmark, in the same way a hypothesis-only baseline is now reported alongside a natural language inference dataset [24].

DATA AND CODE AVAILABILITY

Both corpora are public. Every number in this paper is produced by a committed script that writes a JSON file, and no number was transcribed from a session log. The partition is built once by a script that asserts no duplicate-content group crosses a boundary, and is reused unchanged by every model and every training seed. A single notebook reproduces the preprocessing, both transformers at their deployed configurations and the ensemble, reading each configuration from the checkpoint record at runtime rather than from a value typed into the notebook. Code is released for reproduction. The link is withheld for blind review.

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